III. System Model and Problem Formulation

A. System Model

1) Network Scenario: We consider a general multi-beam LEO satellite communication system under NGSO–GSO coexistence. Since the LEO downlink and the GSO system may operate in the same frequency band, the GSO ground station must be protected from excessive co-channel interference. Hereafter, this ground station is referred to as the GS. A OneWeb-like constellation with 16 fixed spot beams per satellite, as shown in Fig. 1, is used as a representative simulation setting, but the proposed framework can be applied to other LEO systems with predictable beam footprints and coverage overlap.

Let $\mathcal{S}(t)$ denote the set of visible LEO satellites at time slot t, and let $\mathcal{B}_s$ denote the set of beams on satellite s. In the adopted simulation setting, each satellite has 16 beams, while the formulation can be extended to systems with different beam numbers.

Each satellite employs a fixed multi-beam layout. For beam b on satellite s, the ground footprint at time slot t is denoted by $\mathcal{F}_{s,b}(t)$. If user u lies in $\mathcal{F}_{s,b}(t)$, beam b on satellite s is regarded as a feasible serving candidate for that user. When multiple satellite–beam pairs cover the same ground area, the overlap provides possible serving alternatives through user association and resource coordination.

When a visible LEO satellite appears close to the GS antenna boresight or forms a near-inline geometry with the desired GSO satellite direction, its downlink transmission may produce a high aggregate EPFD at the GS. The LEO satellite that contributes the largest EPFD risk at time slot t is defined as the *critical satellite*, denoted by $s^{\mathrm{C}}(t)$.

We further consider the along-track north and south neighboring satellites of the critical satellite as candidate helper satellites for subsequent service transfer and relaying, denoted by $s^{\mathrm{N}}(t)$ and $s^{\mathrm{S}}(t)$, respectively, as shown in Fig. 2. Since LEO satellite motion and beam footprints are predictable, the coverage overlap between critical beams and helper beams can be estimated in advance, enabling helper beams to relay users affected by critical-beam shutdown.

2) User Association and Service Model: Let $\mathcal{U}(t)$ denote the set of terrestrial users considered at time slot t, where each user is covered by at least one visible LEO beam. Each user can be associated with at most one satellite–beam pair in the same time slot. The user–beam association indicator $x_{u,s,b}(t)$ is defined as

$$x_{u,s,b}(t) = \begin{cases} 1, & \text{if user } u \text{ is served by beam } b \text{ of satellite } s, \\ 0, & \text{otherwise,} \end{cases}$$

for all $u \in \mathcal{U}(t)$, $s \in \mathcal{S}(t)$, and $b \in \mathcal{B}_s$.

For beam b on satellite s, the number of users served by this beam is

$$N_{s,b}(t) = \sum_{u \in \mathcal{U}(t)} x_{u,s,b}(t). \tag{1}$$

This value is used in the TDMA-based rate model, where users served by the same beam share the available downlink resource.

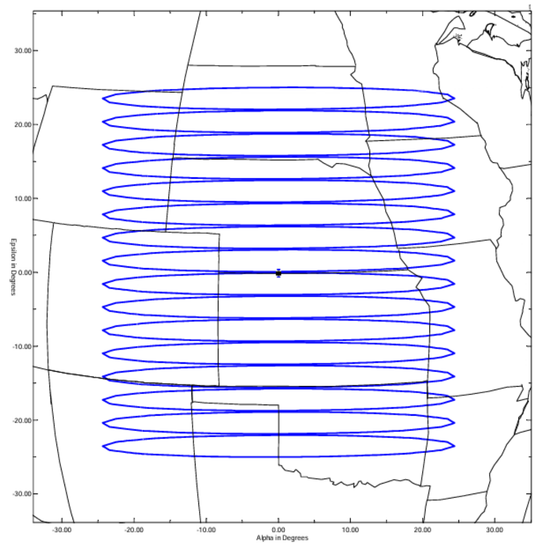


Fig. 1. Ground footprints of the OneWeb Ku-band 16-beams

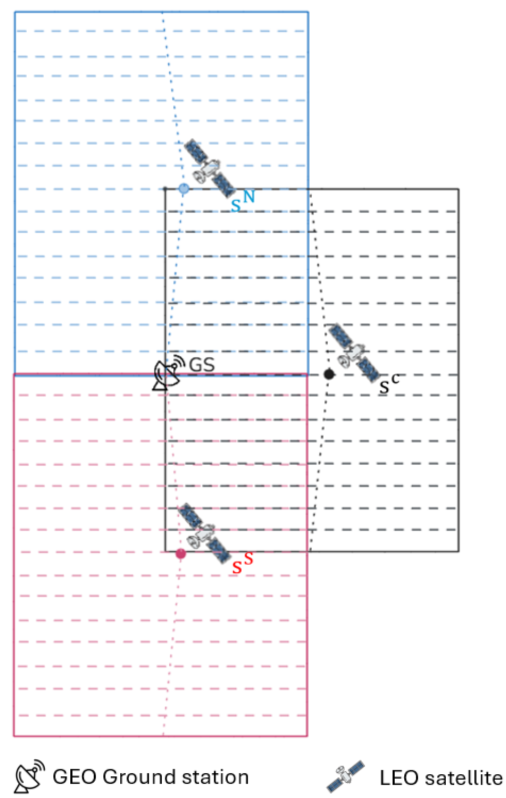


Fig. 2. Network scenario with the protected GSO ground station GS, critical satellite s^{C}, and north/south helper satellites s^{N} and s^{S}.

An association is feasible only when user u lies within the footprint $\mathcal{F}_{s,b}(t)$ of beam b on satellite s. Therefore, association, reassociation, and relay decisions are made only among satellite–beam pairs that can actually cover the user. In overlapping coverage areas, a user may be covered by multiple satellite–beam pairs. Under normal operation, these overlapping beams can coordinate the service demand through user association, time-domain scheduling, or spatial separation, while each user is still associated with only one serving beam in each time slot.

When a critical beam is shut down to satisfy the GS EPFD constraint, the users originally served by this beam can no longer be supported by the critical satellite. If these users are also covered by nearby helper beams, they can be reassociated to feasible helper relay beams. Since the helper relay beams use a compatible downlink service configuration, such as the same service band and polarization setting, the reassociation can be described within the same service model.

3) Beam Power Model: We adopt a default beam-power operation model. For beam b on satellite s, the transmit power at time slot t is denoted by $P_{s,b}(t)$. Under normal operation, when LEO satellites are far from the GS and the EPFD violation risk is low, each active beam operates with its preconfigured default transmit power, denoted by $P_{s,b}^{\mathrm{def}}$.

Different beams are not required to use the same default transmit power. The default power of each beam is preconfigured based on its link condition, so that users in different beam footprints can receive a similar service level under normal operation. Therefore,

$$P_{s,b}(t) = P_{s,b}^{\mathrm{def}}, \qquad \forall s \in \mathcal{S}(t),\ \forall b \in \mathcal{B}_s.$$

This default beam-power profile serves as the normal operating state. If the GS EPFD constraint cannot be satisfied under this state, some beam powers may be adjusted or shut down.

Each beam has a maximum transmit power $P_{s,b}^{\max}$, and

$$0 \le P_{s,b}(t) \le P_{s,b}^{\max}, \qquad \forall s \in \mathcal{S}(t),\ \forall b \in \mathcal{B}_s. \tag{2}$$

Each satellite also has a total transmit power budget $P_s^{\max}$:

$$\sum_{b \in \mathcal{B}_s} P_{s,b}(t) \le P_s^{\max}, \qquad \forall s \in \mathcal{S}(t). \tag{3}$$

When the critical satellite $s^{\mathrm{C}}(t)$ creates an aggregate EPFD violation risk at the GS, some critical beams may be shut down. Let $\mathcal{B}_{\mathrm{C}}^{\mathrm{off}}(t)$ denote the set of beams shut down on the critical satellite, also referred to as closed beams. Then

$$P_{s^{\mathrm{C}},b}(t) = 0, \qquad \forall b \in \mathcal{B}_{\mathrm{C}}^{\mathrm{off}}(t).$$

The remaining active beams on the critical satellite are referred to as safe beams and are denoted by $\mathcal{B}_{\mathrm{C}}^{\mathrm{safe}}(t)$, where

$$\mathcal{B}_{\mathrm{C}}^{\mathrm{safe}}(t) = \mathcal{B}_{s^{\mathrm{C}}} \setminus \mathcal{B}_{\mathrm{C}}^{\mathrm{off}}(t).$$

These safe beams remain active with their default transmit powers:

$$P_{s^{\mathrm{C}},b}(t) = P_{s^{\mathrm{C}},b}^{\mathrm{def}}, \qquad \forall b \in \mathcal{B}_{\mathrm{C}}^{\mathrm{safe}}(t).$$

Hence, the transmit power of a beam on the critical satellite can be written as

$$P_{s^{\mathrm{C}},b}(t) = \begin{cases} 0, & b \in \mathcal{B}_{\mathrm{C}}^{\mathrm{off}}(t), \\ P_{s^{\mathrm{C}},b}^{\mathrm{def}}, & b \in \mathcal{B}_{\mathrm{C}}^{\mathrm{safe}}(t). \end{cases}$$

For helper satellites, beams also operate with their default transmit powers under normal operation. If users affected by critical-beam shutdown are reassociated to feasible helper beams, the transmit power of those helper beams may be adjusted above their default values, subject to the beam-level maximum power and the total satellite power budget (3):

$$P_{s^{\mathrm{h}},b}(t) \le P_{s^{\mathrm{h}},b}^{\max}, \qquad s^{\mathrm{h}} \in \{s^{\mathrm{N}}(t), s^{\mathrm{S}}(t)\}.$$

Throughout this model, each user is still served by only one satellite–beam pair, and the service rate is determined by the selected serving beam and the users sharing that beam.

4) SINR Model: The downlink SINR of a user is modeled using a commonly adopted wireless and satellite communication formulation, where the received signal, co-channel interference, and receiver noise are considered [15]. Let $k_{s,b}$ denote the downlink channel used by beam b on satellite s. Only beams using the same downlink channel as the serving beam are included as co-channel interferers.

If user u is served by beam b on satellite s, its downlink SINR at time slot t is

$$\mathrm{SINR}_u(t) = \frac{S_u(t)}{I_u^{\mathrm{intra}}(t) + I_u^{\mathrm{inter}}(t) + N_u(t)}. \tag{4}$$

where $S_u(t)$ is the desired signal power, $I_u^{\mathrm{intra}}(t)$ is the intra-satellite co-channel interference, $I_u^{\mathrm{inter}}(t)$ is the inter-satellite co-channel interference, and $N_u(t)$ is the receiver noise power.

Desired Signal Power: The desired signal power follows the standard satellite link-budget relation, and the propagation loss components are modeled according to ITU-R recommendations [9], [10]. For user u served by beam b on satellite s,

$$S_u(t) = \frac{P_{s,b}(t)\, G_{s,b\to u}^{\mathrm{tx}}(t)\, G_u^{\mathrm{rx}}}{L_{s,u}(t)}.$$

where $P_{s,b}(t)$ is the transmit power of the serving beam, $G_{s,b\to u}^{\mathrm{tx}}(t)$ is the transmit gain of beam b toward user u, G_u^{rx} is the user receive gain, and $L_{s,u}(t)$ is the total propagation loss between satellite s and user u.

The total propagation loss includes free-space path loss and atmospheric gaseous loss:

$$L_{s,u}(t) = L_{s,u}^{\mathrm{fs}}(t)\, L_{s,u}^{\mathrm{gas}}(t),$$

where $L_{s,u}^{\mathrm{fs}}(t)$ is the free-space path loss and $L_{s,u}^{\mathrm{gas}}(t)$ is the atmospheric gaseous loss.

The free-space path loss follows the standard free-space attenuation model in ITU-R P.525 [9]:

$$L_{s,u}^{\mathrm{fs}}(t) = \left(\frac{4\pi d_{s,u}(t)}{\lambda}\right)^2,$$

where $d_{s,u}(t)$ is the distance between satellite s and user u, and λ is the carrier wavelength.

Intra-Satellite Interference: The intra-satellite interference comes from co-channel beams on the same satellite. Let $\mathcal{B}_s^{\mathrm{co}}(b)$ denote the set of beams on satellite s that reuse the same channel as the serving beam b, excluding beam b itself: The intra-satellite interference is given by

$$I_u^{\mathrm{intra}}(t) = \sum_{b' \in \mathcal{B}_s^{\mathrm{co}}(b)} \frac{P_{s,b'}(t)\, G_{s,b'\to u}^{\mathrm{tx}}(t)\, G_u^{\mathrm{rx}}}{L_{s,u}(t)}.$$

Inter-Satellite Interference: The inter-satellite interference comes from co-channel beams on other visible satellites. Let $\mathcal{B}_{s'}^{\mathrm{co}}(b)$ denote the set of beams on satellite s' that reuse the same channel as the serving beam b: The inter-satellite interference is given by

$$I_u^{\text{inter}}(t) = \sum_{\substack{s' \in \mathcal{S}(t) \\ s' \neq s}} \sum_{b' \in \mathcal{B}_{s'}^{\text{co}}(b)} \frac{P_{s',b'}(t)\, G^{\text{tx}}_{s',b' \to u}(t)\, G^{\text{rx}}_u}{L_{s',u}(t)}.$$

where $G^{\text{tx}}_{s',b' \to u}(t)$ is the transmit gain of the interfering beam toward user u, and $L_{s',u}(t)$ is the total propagation loss between satellite s' and user u.

With this commonly used downlink SINR model, the user SINR changes when the serving beam, beam transmit power, propagation loss, or co-channel interference changes.

5) Capacity and User Satisfaction Model: After obtaining the user SINR, the instantaneous link capacity is modeled by the Shannon capacity formula, which is commonly used in wireless and satellite communication systems [16], [17]. Let B_u denote the effective bandwidth allocated to user u. The instantaneous capacity of user u at time slot t is

$$C_u(t) = B_u \log_2\!\big(1 + \text{SINR}_u(t)\big).$$

When multiple users are served by the same beam, they share the downlink resource of that beam. Following the time-division resource sharing model commonly used in satellite downlink systems [17], [18], the beam resource is assumed to be equally divided among the users served by the same beam. The actual service rate of user u is therefore

$$R_u(t) = \frac{C_u(t)}{N_{s,b}(t)}. \tag{5}$$

where $N_{s,b}(t)$ is defined in (1). This expression represents an equal-time sharing model, where users served by the same beam are allocated the same fraction of the beam resource in each time slot.

Let $D_u(t)$ denote the demand rate of user u at time slot t. The satisfaction of user u is defined as

$$\text{sat}_u(t) = \min\left(\frac{R_u(t)}{D_u(t)}, 1\right). \tag{6}$$

where $\text{sat}_u(t) \in [0,1]$. If $\text{sat}_u(t) = 1$, the demand of user u is fully met; if $0 < \text{sat}_u(t) < 1$, user u receives partial service; if $\text{sat}_u(t) = 0$, user u receives no effective service in the time slot.

This satisfaction metric represents the fraction of user demand that is satisfied in each time slot.

6) Beam-Level EPFD Model: The EPFD calculation follows the ITU Radio Regulations and ITU-R S.1503 framework for NGSO–GSO coexistence [5], [6]. Since different beams on the same LEO satellite have different pointing directions, antenna gains, and transmit powers toward the GS, the EPFD contribution is evaluated at the beam level and summed in the linear domain.

For EPFD calculation, let $P^{\text{ref}}_{s,b}(t)$ denote the radio-frequency (RF) transmit power of beam b on satellite s within the ITU reference bandwidth B_{ref} specified for the applicable EPFD limit [7], [8]. If $P_{s,b}(t)$ is defined over a wider beam bandwidth, it is converted to the corresponding power within B_{ref} before applying the EPFD formula. The beam-level EPFD contribution toward the GS is given by

$$\text{EPFD}_{s,b}(t) = \frac{P^{\text{ref}}_{s,b}(t)\, G^{\text{tx}}_{s,b\to g}(t)}{4\pi d^2_{s,g}(t)} \cdot \frac{G^{\text{rx}}_g(\theta_{g\to s}(t))}{G^{\text{rx}}_{g,\max}},$$

where $G^{\text{tx}}_{s,b\to g}(t)$ is the transmit antenna gain of beam b on satellite s in the direction of the GS, $d_{s,g}(t)$ is the distance between satellite s and the GS, $G^{\text{rx}}_g(\theta_{g\to s}(t))$ is the receive antenna gain of the reference GS antenna in the direction of satellite s, $G^{\text{rx}}_{g,\max}$ is the maximum receive antenna gain of the reference GS antenna, and $\theta_{g\to s}(t)$ is the off-axis angle between the GS antenna boresight and the direction of satellite s.

The transmit antenna gain $G^{\text{tx}}_{s,b\to g}(t)$ is evaluated from the beam antenna pattern according to the off-axis angle between the boresight of beam b and the direction of the GS. Since each beam has a different pointing direction, the EPFD contribution of each beam is computed individually.

The aggregate EPFD at the GS is obtained by summing the beam-level EPFD contributions from all beams of all visible LEO satellites:

$$\text{EPFD}_{\text{agg}}(t) = \sum_{s\in\mathcal{S}(t)} \sum_{b\in\mathcal{B}_s} \text{EPFD}_{s,b}(t).$$

The resulting aggregate EPFD is compared with the applicable ITU EPFD limit in the problem formulation.

B. Problem Formulation

Based on the system model above, we formulate the user association and beam power control problem to maximize the total user satisfaction while satisfying association, power, and EPFD constraints. The optimization variables are the user–beam association indicators $\mathbf{x}(t) = \{x_{u,s,b}(t)\}$ and the beam transmit powers $\mathbf{P}(t) = \{P_{s,b}(t)\}$.

Objective function:

$$\max_{\mathbf{x}(t),\mathbf{P}(t)} \sum_{u\in\mathcal{U}(t)} \text{sat}_u(t). \tag{7}$$

Constraints:

$$\sum_{s\in\mathcal{S}(t)} \sum_{b\in\mathcal{B}_s} x_{u,s,b}(t) \le 1, \quad \forall u \in \mathcal{U}(t), \tag{8}$$

$$x_{u,s,b}(t) \in \{0,1\}, \quad \forall u \in \mathcal{U}(t),\ \forall s \in \mathcal{S}(t),\ \forall b \in \mathcal{B}_s, \tag{9}$$

$$0 \le P_{s,b}(t) \le P^{\max}_{s,b}, \quad \forall s \in \mathcal{S}(t),\ \forall b \in \mathcal{B}_s. \tag{10}$$

$$\sum_{b\in\mathcal{B}_s} P_{s,b}(t) \le P^{\max}_s, \quad \forall s \in \mathcal{S}(t). \tag{11}$$

$$\text{EPFD}_{\text{agg}}(t) \le \text{EPFD}_{\text{lim}}. \tag{12}$$

Constraint (8) ensures that each user is associated with at most one satellite–beam pair in each time slot.

Constraint (9) defines the binary association decision, where $x_{u,s,b}(t)$ can only be 0 or 1 and therefore cannot take a negative value.

Constraint (10) limits the transmit power of each beam, including the possibility of beam shutdown when $P_{s,b}(t) = 0$.

Constraint (11) enforces the total transmit power budget of each satellite.

Constraint (12) ensures that the aggregate EPFD at the GS does not exceed the applicable ITU limit.

The objective in (7) maximizes the aggregate satisfaction of all users in the considered time slot. Although the objective is written as a summation of user satisfaction, each satisfaction value depends on the selected serving beam, beam loading, SINR, and service rate. In addition, beam power affects both the user-side downlink performance and the GS-side aggregate EPFD. Therefore, the association and power decisions are strongly coupled.

In critical GS geometries, reducing or shutting down beam power can lower the aggregate EPFD, but it may also reduce the service rate of users originally served by the affected beams. These users can be reassociated to feasible helper beams, but doing so changes the helper beam loading and may further affect the service rates of other users.

Because of these coupled association, power, SINR, service-rate, and EPFD relationships, the problem becomes a mixed-integer nonlinear optimization problem. Solving the global optimum in every time slot is computationally difficult. Therefore, the next chapter develops a staged heuristic framework to obtain feasible association and power decisions during critical periods.

IV. Method

A. Method Overview

Chapter III formulates the considered problem as a joint user association and beam power control problem under user association, beam power, satellite power budget, and EPFD constraints. The association variables are discrete, while the beam power variables are continuous, making the problem a mixed-integer nonlinear optimization problem. In addition, user association affects beam loading, downlink SINR, time-division resource sharing, and satisfaction, while beam power affects both user service quality and the aggregate EPFD at the GS. Directly solving the global optimum for every time slot is therefore computationally demanding.

To reduce online decision complexity, we propose Safe-Beam Assisted Power-Release Relay (SAPR-R). SAPR-R uses the predictable LEO satellite motion, GS location, GSO direction, and beam footprints to precompute beam-level relation records for all time slots in the observation window. For each time slot, the record includes the critical satellite, the north and south helper satellites, the closed beams on the critical satellite, the safe beams that remain active with default power, and the beam-pair mappings between critical beams and helper beams. This precomputation depends only on satellite geometry, beam footprints, and beam-level EPFD calculation, rather than user distribution, and can therefore be completed before online user reassociation.

During online operation, SAPR-R loads the beam relation record of the current time slot and makes user-aware reassociation decisions based on user locations, current associations, and service satisfaction. First, Safe-Beam Assisted Helper Relief (SAHR) lets critical safe beams take over selected users originally served by helper release beams. A reassociation is accepted only when it does not reduce the satisfaction of the transferred user or the users already served by the critical safe beam. The transmit power no longer required by the helper release beams is then added to the available power budget of the helper satellite.

Next, the helper satellite allocates its available power budget to helper relay beams according to relay demand, so that these beams can provide additional service capacity for users affected by closed beams. Finally, Beam-Pair Local Relay (BPLR) reassociates closed-beam users to feasible helper relay beams within the precomputed beam-pair mappings. Candidate users are served sequentially according to relay link quality and remaining relay capacity.

In summary, SAPR-R consists of the following steps:

- **Time-slot beam relation record:** Precompute the critical/helper satellites, closed/safe beams, and beam-pair mappings for each time slot.
- **Safe-Beam Assisted Helper Relief (SAHR):** Use critical safe beams to take over selected helper users without reducing satisfaction, thereby releasing helper satellite power.
- **Helper relay beam boosting:** Allocate the available helper satellite power budget to helper relay beams according to relay demand.
- **Beam-Pair Local Relay (BPLR):** Reassociate closed-beam users to feasible helper relay beams within each beam pair according to relay link quality and remaining relay capacity.

B. Time-Slot Beam Relation Record

For each time slot t, SAPR-R constructs a beam relation record $\mathcal{R}_B(t)$, which stores the beam-level information used by the online reassociation and relay procedures. Figure 3 illustrates the time-varying beam status of the critical satellite and its helper satellites. The record is defined as

$$\begin{aligned}\mathcal{R}_B(t) = \big(&s^{\mathrm{C}}(t),\, s^{\mathrm{N}}(t),\, s^{\mathrm{S}}(t),\\ &\mathcal{B}_c^{\mathrm{off}}(t),\, \mathcal{B}_c^{\mathrm{safe}}(t),\\ &\mathcal{M}_{\mathrm{safe}\to\mathrm{rel}}(t),\, \mathcal{M}_{\mathrm{off}\to\mathrm{relay}}(t)\big).\end{aligned}$$

In this record, $s^{\mathrm{C}}(t)$, $s^{\mathrm{N}}(t)$, and $s^{\mathrm{S}}(t)$ denote the critical satellite, north helper satellite, and south helper satellite, respectively. $\mathcal{B}_c^{\mathrm{off}}(t)$ denotes the set of closed beams on the critical satellite, and $\mathcal{B}_c^{\mathrm{safe}}(t)$ denotes the set of safe beams that remain active with default power. $\mathcal{M}_{\mathrm{safe}\to\mathrm{rel}}(t)$ and $\mathcal{M}_{\mathrm{off}\to\mathrm{relay}}(t)$ denote the safe-beam-to-helper-release-beam mapping and the closed-beam-to-helper-relay-beam mapping, respectively.

1) Critical and Helper Satellite Identification: In each time slot t, the critical satellite $s^{\mathrm{C}}(t)$ is identified as the visible LEO satellite that contributes the largest EPFD risk at the GS. Since the EPFD violation is mainly caused by the critical satellite, beam shutdown is applied only to beams on $s^{\mathrm{C}}(t)$, while helper and other visible satellites remain available for service continuity. The along-track north and south neighboring satellites of $s^{\mathrm{C}}(t)$ are selected as helper satellites, denoted by $s^{\mathrm{N}}(t)$ and $s^{\mathrm{S}}(t)$, respectively.

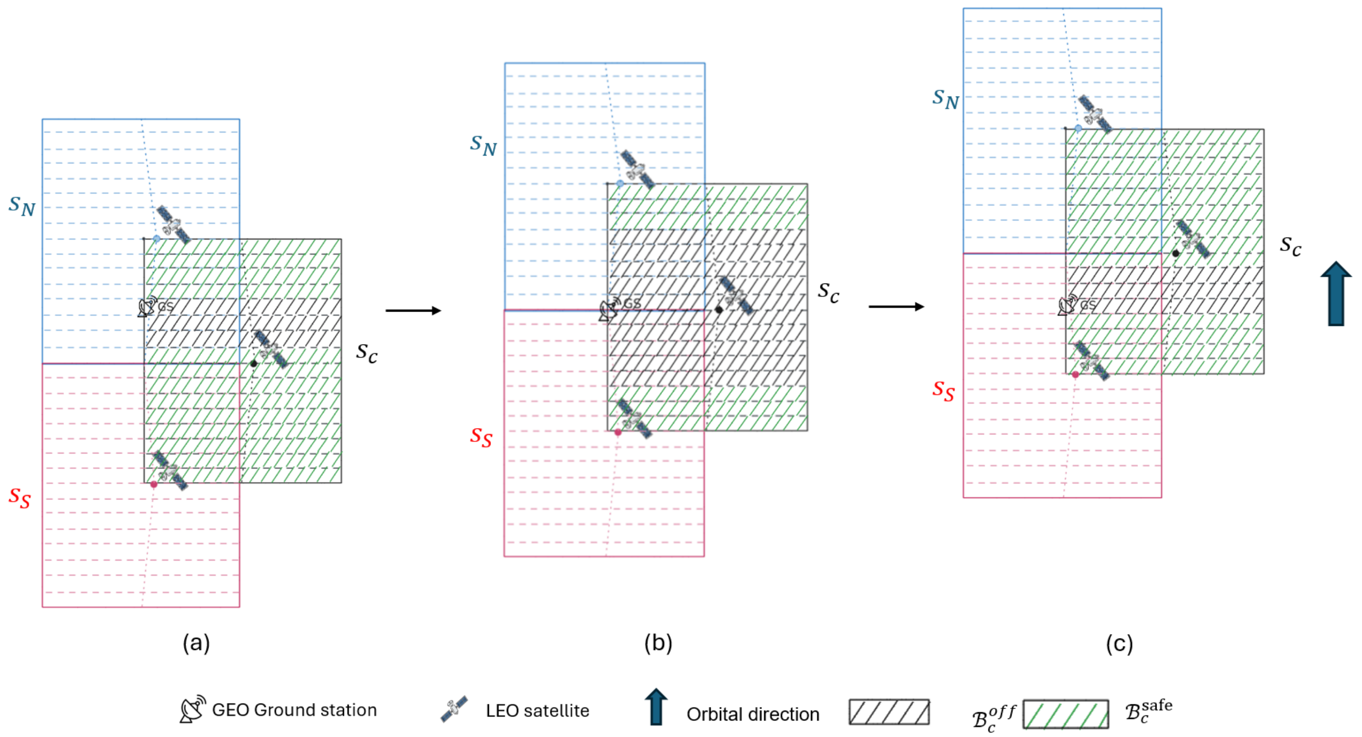


Fig. 3. Three time slots as the along-track LEO satellites (s^{N}, s^{c}, and s^{S}) pass the protected GSO ground station. In region S_{C} on the critical satellite, black-hatched beams are $\mathcal{B}_c^{\text{off}}(t)$ (closed beams) and green-hatched beams are $\mathcal{B}_c^{\text{safe}}(t)$ (safe beams). Panels (a)–(c) show earlier to later time slots.

2) EPFD-Driven Beam Status Determination: For each beam b on the critical satellite, the beam-level EPFD risk score is defined as

$$q_{c,b}(t) = \mathrm{EPFD}_{s^{\mathrm{C}},b}(t).$$

The critical beams are sorted in descending order of $q_{c,b}(t)$. A greedy shutdown rule is then applied: at each step, the beam with the highest EPFD risk score is shut down until the aggregate EPFD at the GS satisfies the applicable EPFD limit.

The beams shut down by this procedure form the closed-beam set $\mathcal{B}_c^{\text{off}}(t)$, and the remaining active beams form the safe-beam set:

$$\mathcal{B}_c^{\text{safe}}(t) = \mathcal{B}_{s^{\mathrm{C}}} \setminus \mathcal{B}_c^{\text{off}}(t).$$

Because this beam status determination depends only on beam-level EPFD contributions and the EPFD limit, it is independent of user distribution and can be precomputed before online user reassociation.

3) Beam-Pair Mapping Construction: Given the closed-beam set $\mathcal{B}_c^{\text{off}}(t)$ and the safe-beam set $\mathcal{B}_c^{\text{safe}}(t)$, SAPR-R constructs beam-pair mappings based on footprint overlap between the critical satellite and its helper satellites.

For SAHR, the safe-beam-to-helper-release-beam mapping is defined as

$$\begin{aligned}\mathcal{M}_{\text{safe}\to\text{rel}}(t) = \big\{(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}) :\;& \\ & b_c^{\text{safe}} \in \mathcal{B}_c^{\text{safe}}(t), \\ & s_h \in \{s^{\mathrm{N}}(t), s^{\mathrm{S}}(t)\}, \\ & b_h^{\text{rel}} \in \mathcal{B}_{s_h}, \\ & \mathcal{F}_{s^{\mathrm{C}},b_c^{\text{safe}}}(t) \cap \mathcal{F}_{s_h,b_h^{\text{rel}}}(t) \neq \emptyset\big\}.\end{aligned}$$

This mapping identifies the helper beams whose users can be reassociated to critical safe beams.

For BPLR, the closed-beam-to-helper-relay-beam mapping is defined as

$$\begin{aligned}\mathcal{M}_{\text{off}\to\text{relay}}(t) = \big\{(b_c^{\text{off}}, s_h, b_h^{\text{relay}}) :\;& \\ & b_c^{\text{off}} \in \mathcal{B}_c^{\text{off}}(t), \\ & s_h \in \{s^{\mathrm{N}}(t), s^{\mathrm{S}}(t)\}, \\ & b_h^{\text{relay}} \in \mathcal{B}_{s_h}, \\ & \mathcal{F}_{s^{\mathrm{C}},b_c^{\text{off}}}(t) \cap \mathcal{F}_{s_h,b_h^{\text{relay}}}(t) \neq \emptyset\big\}.\end{aligned}$$

This mapping identifies the helper beams that can serve users originally located under critical closed beams.

Figure 4 illustrates the two mappings stored in $\mathcal{R}_B(t)$: $\mathcal{M}_{\text{safe}\to\text{rel}}(t)$ for SAHR and $\mathcal{M}_{\text{off}\to\text{relay}}(t)$ for BPLR. These mappings rely only on footprint overlap and therefore can be applied to general multi-beam LEO systems, as long as the beam footprints and helper-satellite geometry are known. In the considered along-track helper scenario, the overlap is local and can be handled within predefined beam pairs, avoiding a global user-to-beam matching process.

For each helper satellite, helper beams are assigned to one of two roles in a time slot: release beams for SAHR and relay beams for BPLR. To avoid role conflict, the two sets are kept disjoint:

$$\mathcal{B}_{s_h}^{\text{rel}}(t) \cap \mathcal{B}_{s_h}^{\text{relay}}(t) = \emptyset, \qquad \forall s_h \in \{s^{\mathrm{N}}(t), s^{\mathrm{S}}(t)\}.$$

C. Safe-Beam Assisted Helper Relief

After loading $\mathcal{R}_B(t)$, the online procedure starts with SAHR, as illustrated in Fig. 5. SAHR does not serve critical

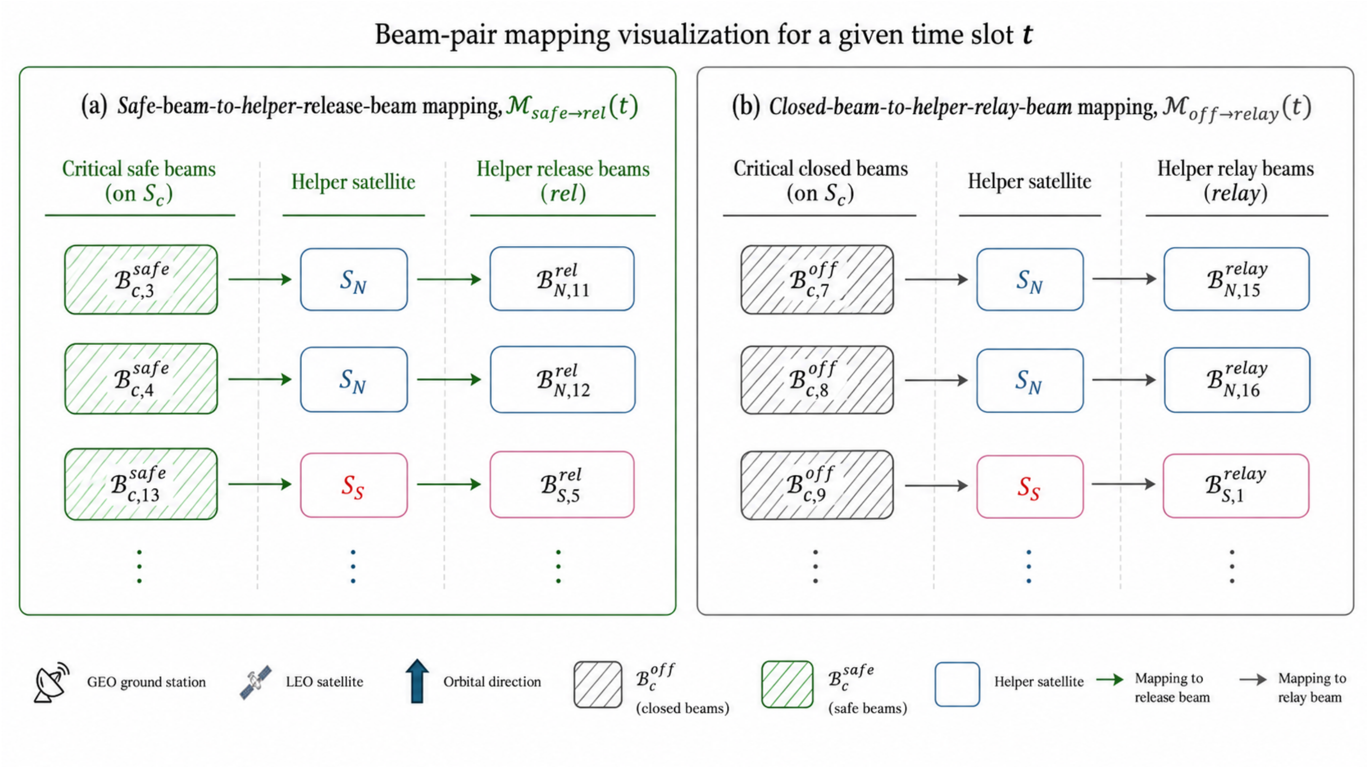


Fig. 4. Beam-pair mapping tables at time slot t. (a) $\mathcal{M}_{\text{safe}\to\text{rel}}(t)$: critical safe beams on s^{C} mapped to helper release beams on $s^{\text{N}}/s^{\text{S}}$ (SAHR). (b) $\mathcal{M}_{\text{off}\to\text{relay}}(t)$: critical closed beams mapped to helper relay beams (BPLR).

closed-beam users directly. Instead, it uses critical safe beams to take over selected users originally served by helper release beams, so that the corresponding helper satellite can make more power available for subsequent relay-beam boosting.

For each mapping tuple $(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}) \in \mathcal{M}_{\text{safe}\to\text{rel}}(t)$, SAPR-R identifies the helper users that can be transferred from the helper release beam to the critical safe beam. Let u_h denote a helper user. The candidate helper-user set is defined as

$$\begin{aligned}&\mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}, t) = \\ &\quad\Big\{u_h \in \mathcal{U}(t) : x_{u_h, s_h, b_h^{\text{rel}}}(t) = 1, \\ &\qquad u_h \in \mathcal{F}_{s^{\text{C}}, b_c^{\text{safe}}}(t)\Big\}.\end{aligned}$$

The first condition means that user u_h is currently served by helper release beam b_h^{rel} on helper satellite s_h. The second condition means that the same user is also covered by the critical safe beam b_c^{safe}. Therefore, users in $\mathcal{V}_{\text{rel}}(b_c^{\text{safe}}, s_h, b_h^{\text{rel}}, t)$ are eligible to be reassociated from the helper release beam to the critical safe beam.

If candidate helper user u_h is reassociated to b_c^{safe}, the reduction in the required transmit power of the helper release beam is defined as

$$\Delta P_{u_h}^{\text{rel}}(t) = P_{b_h^{\text{rel}}}^{\text{req,old}}(t) - P_{b_h^{\text{rel}}}^{\text{req,new}}(u_h, t).$$

Here, $P_{b_h^{\text{rel}}}^{\text{req,old}}(t)$ denotes the required transmit power of helper release beam b_h^{rel} before transferring u_h, while $P_{b_h^{\text{rel}}}^{\text{req,new}}(u_h, t)$ denotes the required transmit power of the same beam after u_h is removed from that beam. Therefore, $\Delta P_{u_h}^{\text{rel}}(t)$ represents the amount of helper transmit power released by transferring u_h to the critical safe beam.

Candidates are ranked in descending order of $\Delta P_{u_h}^{\text{rel}}(t)$, so users that release more helper power are considered first. A trial reassociation is accepted only when the transferred helper user and the original users on the critical safe beam do not lose satisfaction:

$$\text{sat}_{u_h}^{\text{new}}(t) \ge \text{sat}_{u_h}^{\text{old}}(t),$$

$$\text{sat}_{u_s}^{\text{new}}(t) \ge \text{sat}_{u_s}^{\text{old}}(t), \qquad \forall u_s \in \mathcal{U}_{b_c^{\text{safe}}}^{\text{ori}}(t).$$

Accepted transfers form the set $\mathcal{A}_{\text{rel}}(s_h, t)$. The total released power on helper satellite s_h is

$$P_{s_h}^{\text{rel}}(t) = \sum_{u_h \in \mathcal{A}_{\text{rel}}(s_h, t)} \Delta P_{u_h}^{\text{rel}}(t).$$

The available power budget for relay-beam boosting consists of the unused satellite power under the default beam-power profile and the power released by SAHR:

$$P_{s_h}^{\text{pool}}(t) = P_{s_h}^{\text{unused}}(t) + P_{s_h}^{\text{rel}}(t),$$

where

$$P_{s_h}^{\text{unused}}(t) = P_{s_h}^{\max} - \sum_{b \in \mathcal{B}_{s_h}} P_{s_h, b}^{\text{def}}.$$

Helper beams that are not selected as release beams or relay beams keep their default transmit powers.

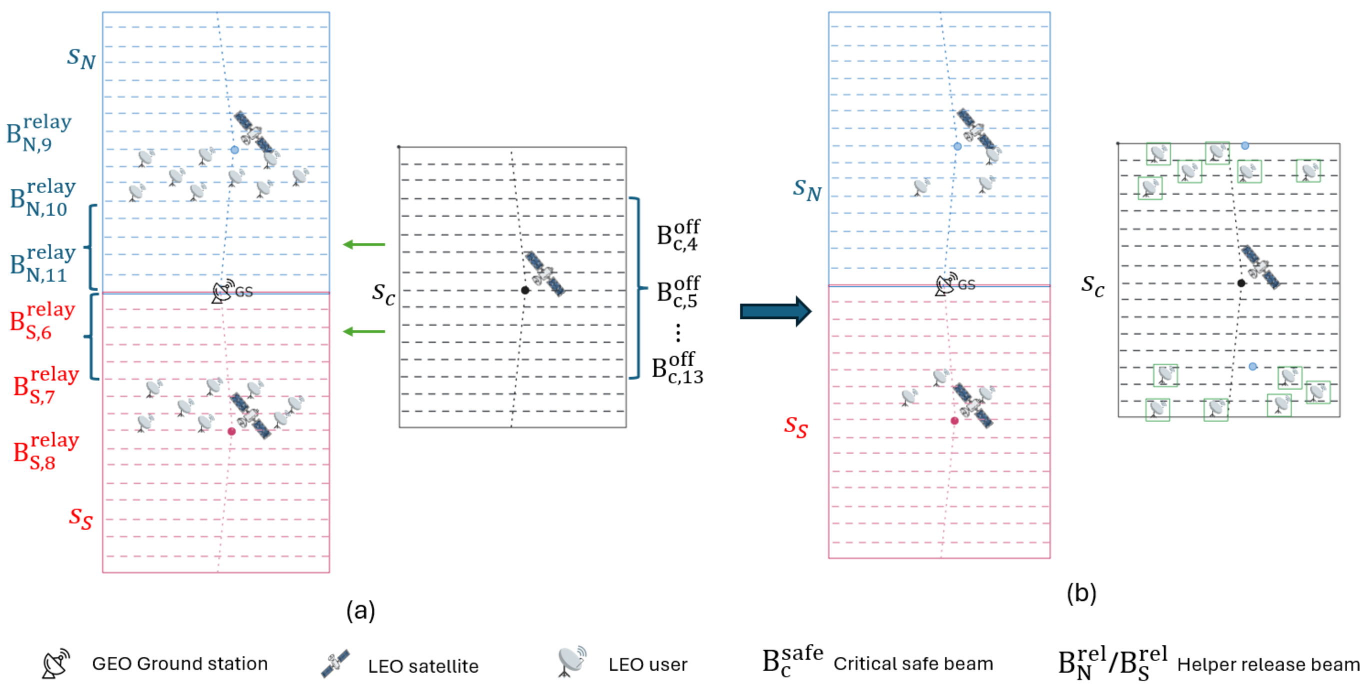


Fig. 5. Illustration of Safe-Beam Assisted Helper Relief (SAHR). Critical safe beams take over eligible helper users from helper release beams, and the released power is returned to the helper satellite power pool.

D. Helper Relay Beam Boosting

After SAHR, each helper satellite s_h uses its available relay power budget $P^{\text{pool}}_{s_h}(t)$, defined in the previous subsection, to increase the transmit power of helper relay beams. For helper satellite s_h, the relay beam set is defined as

$$\mathcal{B}^{\text{relay}}_{s_h}(t) = \big\{ b^{\text{relay}}_h : \exists\, b^{\text{off}}_c \in \mathcal{B}^{\text{off}}_c(t) : (b^{\text{off}}_c, s_h, b^{\text{relay}}_h) \in \mathcal{M}_{\text{off}\to\text{relay}}(t) \big\}.$$

This set contains the helper beams on s_h that are mapped to at least one closed beam on the critical satellite and can therefore be used for BPLR.

To estimate the relay demand of each helper relay beam, SAPR-R first identifies the candidate closed-beam users that can be served by that beam. Let u_c denote a user originally served by a closed beam on the critical satellite. For helper relay beam b^{relay}_h, the candidate user set is defined as

$$\mathcal{U}_{\text{cand}}(b^{\text{relay}}_h, t) = \Big\{ u_c \in \mathcal{U}(t) : \exists\, b^{\text{off}}_c \in \mathcal{B}^{\text{off}}_c(t) : (b^{\text{off}}_c, s_h, b^{\text{relay}}_h) \in \mathcal{M}_{\text{off}\to\text{relay}}(t), \; x_{u_c, s^{\text{C}}, b^{\text{off}}_c}(t) = 1,\; u_c \in \mathcal{F}_{s_h, b^{\text{relay}}_h}(t) \Big\}.$$

The condition $x_{u_c, s^{\text{C}}, b^{\text{off}}_c}(t) = 1$ means that user u_c is originally associated with the closed beam b^{off}_c on the critical satellite. The footprint condition means that the same user is also covered by helper relay beam b^{relay}_h. Therefore, $\mathcal{U}_{\text{cand}}(b^{\text{relay}}_h, t)$ represents the closed-beam users that are eligible to be relayed by b^{relay}_h.

The relay demand of helper relay beam b^{relay}_h is defined as the number of its candidate users:

$$D^{\text{relay}}_{b^{\text{relay}}_h}(t) = \left| \mathcal{U}_{\text{cand}}(b^{\text{relay}}_h, t) \right|.$$

This demand value is used only to prioritize relay-beam boosting. The actual user reassociation is performed later in BPLR after the final transmit power of each helper relay beam is determined.

For a helper relay beam b^{relay}_h, let $\Delta P^{\text{boost,max}}_{b^{\text{relay}}_h}(t)$ denote the additional power required to increase its transmit power from the default value to the beam-level maximum:

$$\Delta P^{\text{boost,max}}_{b^{\text{relay}}_h}(t) = P^{\max}_{s_h, b^{\text{relay}}_h} - P^{\text{def}}_{s_h, b^{\text{relay}}_h}.$$

If $P^{\text{pool}}_{s_h}(t)$ is sufficient to raise all helper relay beams to their beam-level maximum powers, all relay beams are fully boosted. Otherwise, helper relay beams are sorted in descending order of $D^{\text{relay}}_{b^{\text{relay}}_h}(t)$, and beams with higher relay demand receive additional power first.

For each helper relay beam, if the remaining available relay power budget is sufficient, its transmit power is increased to $P^{\max}_{s_h, b^{\text{relay}}_h}$. If the remaining budget is positive but insufficient for full boosting, the remaining power is assigned as a partial boost above $P^{\text{def}}_{s_h, b^{\text{relay}}_h}$. This avoids leaving unused relay power and increases the available capacity for BPLR.

After this step, each helper relay beam proceeds to BPLR with its final transmit power.

E. Beam-Pair Local Relay

After relay-beam boosting, BPLR performs the actual reassociation of closed-beam users to helper relay beams, as illustrated in Fig. 6. For each mapping tuple $(b^{\text{off}}_c, s_h, b^{\text{relay}}_h) \in \mathcal{M}_{\text{off}\to\text{relay}}(t)$, BPLR considers the users in $\mathcal{U}_{\text{cand}}(b^{\text{relay}}_h, t)$ that were originally served by the closed beam b^{off}_c. Since $\mathcal{U}_{\text{cand}}(b^{\text{relay}}_h, t)$ already includes the coverage condition of the helper relay beam, these users are eligible to be reassociated to b_h^{relay}.

Let u_c denote a closed-beam user considered in this beam pair. For each candidate user u_c, let $R_{u_c,b_h^{\mathrm{relay}}}(t)$ denote the achievable service rate if u_c is served by helper relay beam b_h^{relay} using the final transmit power obtained after relay-beam boosting. Candidates are ranked in descending order of $R_{u_c,b_h^{\mathrm{relay}}}(t)$, so users with better relay link quality are considered first.

Let $C^{\mathrm{rem}}_{b_h^{\mathrm{relay}}}(t)$ denote the remaining relay capacity of helper relay beam b_h^{relay} after preserving the service of its original helper users. This remaining capacity is allocated sequentially to the sorted closed-beam users. If $C^{\mathrm{rem}}_{b_h^{\mathrm{relay}}}(t)$ can satisfy the demand $D_{u_c}(t)$ of user u_c, the user is reassociated to b_h^{relay} and its satisfaction is updated. If the remaining capacity is positive but insufficient to fully satisfy the next user, partial capacity is assigned to that user and the beam-pair process stops. When the remaining capacity is exhausted, no further users are reassociated in this beam pair.

All relay decisions follow the single-association constraint. A relayed user is moved from the critical closed beam to the helper relay beam and is served by only one beam in the time slot.

F. Overall Algorithm

The online procedure uses a single time-slot-based algorithm. In each time slot, SAPR-R first runs SAHR and relay-beam boosting, then runs BPLR using the updated helper relay-beam powers. The complete procedure is given in Algorithm 1.

Algorithm 1 SAPR-R Online Operation

Input: $\{\mathcal{R}_B(t)\}_{t=1}^T$, initial $\mathbf{x}(t)$, $\mathcal{U}(t)$, $D_u(t)$
Input: $P^{\mathrm{def}}_{s,b}$, $P^{\mathrm{max}}_{s,b}$, P^{max}_{s}, $\mathcal{F}_{s,b}(t)$, capacity/satisfaction model
Output: $\mathbf{x}^{\mathrm{final}}(t)$, $P^{\mathrm{final}}_{s,b}(t)$, $\mathrm{sat}^{\mathrm{final}}_u(t)$

for $t = 1$ **to** T **do**
  Load $\mathcal{R}_B(t)$; initialize beam powers per default profile and closed/safe beam status
  Initialize $P^{\mathrm{rel}}_{s_h}(t) = 0$ for each $s_h \in \{s^{\mathrm{N}}(t), s^{\mathrm{S}}(t)\}$
  Stage 1: SAHR
  for all $(b_c^{\mathrm{safe}}, s_h, b_h^{\mathrm{rel}}) \in \mathcal{M}_{\mathrm{safe}\to\mathrm{rel}}(t)$ **do**
    Build $\mathcal{V}_{\mathrm{rel}}(b_c^{\mathrm{safe}}, s_h, b_h^{\mathrm{rel}}, t)$ using $\mathbf{x}(t)$ and beam footprints
    Compute $\Delta P^{\mathrm{rel}}_{u_h}(t)$ for each $u_h \in \mathcal{V}_{\mathrm{rel}}(b_c^{\mathrm{safe}}, s_h, b_h^{\mathrm{rel}}, t)$
    Sort candidate helper users by $\Delta P^{\mathrm{rel}}_{u_h}(t)$ descending
    for all u_h in sorted order **do**
      Trial reassociation of u_h from b_h^{rel} to b_c^{safe}
      if transferred helper user and original users on b_c^{safe} do not lose satisfaction **then**
        Accept the reassociation
        Update $\mathbf{x}(t)$, satisfaction values, and $P^{\mathrm{rel}}_{s_h}(t)$
      end if
    end for
  end for
  Stage 1 continued: helper relay beam boosting
  for all $s_h \in \{s^{\mathrm{N}}(t), s^{\mathrm{S}}(t)\}$ **do**
    Compute unused power $P^{\mathrm{unused}}_{s_h}(t)$ under default beam-power profile
    Compute relay power budget $P^{\mathrm{pool}}_{s_h}(t) = P^{\mathrm{unused}}_{s_h}(t) + P^{\mathrm{rel}}_{s_h}(t)$
    Extract helper relay beam set $\mathcal{B}^{\mathrm{relay}}_{s_h}(t)$ from $\mathcal{M}_{\mathrm{off}\to\mathrm{relay}}(t)$
    for all $b_h^{\mathrm{relay}} \in \mathcal{B}^{\mathrm{relay}}_{s_h}(t)$ **do**
      Build $\mathcal{U}_{\mathrm{cand}}(b_h^{\mathrm{relay}}, t)$
      Compute relay demand $D^{\mathrm{relay}}_{b_h^{\mathrm{relay}}}(t) = |\mathcal{U}_{\mathrm{cand}}(b_h^{\mathrm{relay}}, t)|$
      Compute $\Delta P^{\mathrm{boost,max}}_{b_h^{\mathrm{relay}}}(t)$ for boosting b_h^{relay} to beam-level maximum
    end for
    Sort helper relay beams by $D^{\mathrm{relay}}_{b_h^{\mathrm{relay}}}(t)$ descending
    for all b_h^{relay} in sorted order **do**
      Allocate $P^{\mathrm{pool}}_{s_h}(t)$ to b_h^{relay}: full boost, partial boost, or default power
      Update final transmit power of b_h^{relay}
    end for
  end for
  Stage 2: Beam-Pair Local Relay
  for all $(b_c^{\mathrm{off}}, s_h, b_h^{\mathrm{relay}}) \in \mathcal{M}_{\mathrm{off}\to\mathrm{relay}}(t)$ **do**
    Select from $\mathcal{U}_{\mathrm{cand}}(b_h^{\mathrm{relay}}, t)$ users originally served by b_c^{off}
    Compute remaining relay capacity of b_h^{relay} after preserving original helper users
    Compute $R_{u_c,b_h^{\mathrm{relay}}}(t)$ for each candidate u_c using final power of b_h^{relay}
    Sort candidate closed-beam users by $R_{u_c,b_h^{\mathrm{relay}}}(t)$ descending
    for all u_c in sorted order **do**
      if remaining relay capacity can satisfy $D_{u_c}(t)$ **then**
        Reassociate u_c to b_h^{relay}
        Update $\mathrm{sat}_{u_c}(t)$ and remaining relay capacity
      else if remaining relay capacity > 0 **then**
        Assign partial capacity to u_c
        Update $\mathrm{sat}_{u_c}(t)$; set remaining relay capacity to 0; **break**
      else
        break
      end if
    end for
  end for
  Validate single-association, beam power, satellite total power, and EPFD constraints
  Store $\mathbf{x}^{\mathrm{final}}(t)$, $P^{\mathrm{final}}_{s,b}(t)$, and $\mathrm{sat}^{\mathrm{final}}_u(t)$
end for

G. Complexity Discussion

SAPR-R avoids directly solving the mixed-integer nonlinear optimization problem formulated in Chapter III. Since the beam relation record $\mathcal{R}_B(t)$ is precomputed, the online procedure does not need to re-identify the critical satellite, helper satellites, or beam-pair mappings in each time slot. The online computation is therefore limited to candidate construction, local sorting, power allocation, and sequential reassociation within predefined beam pairs.

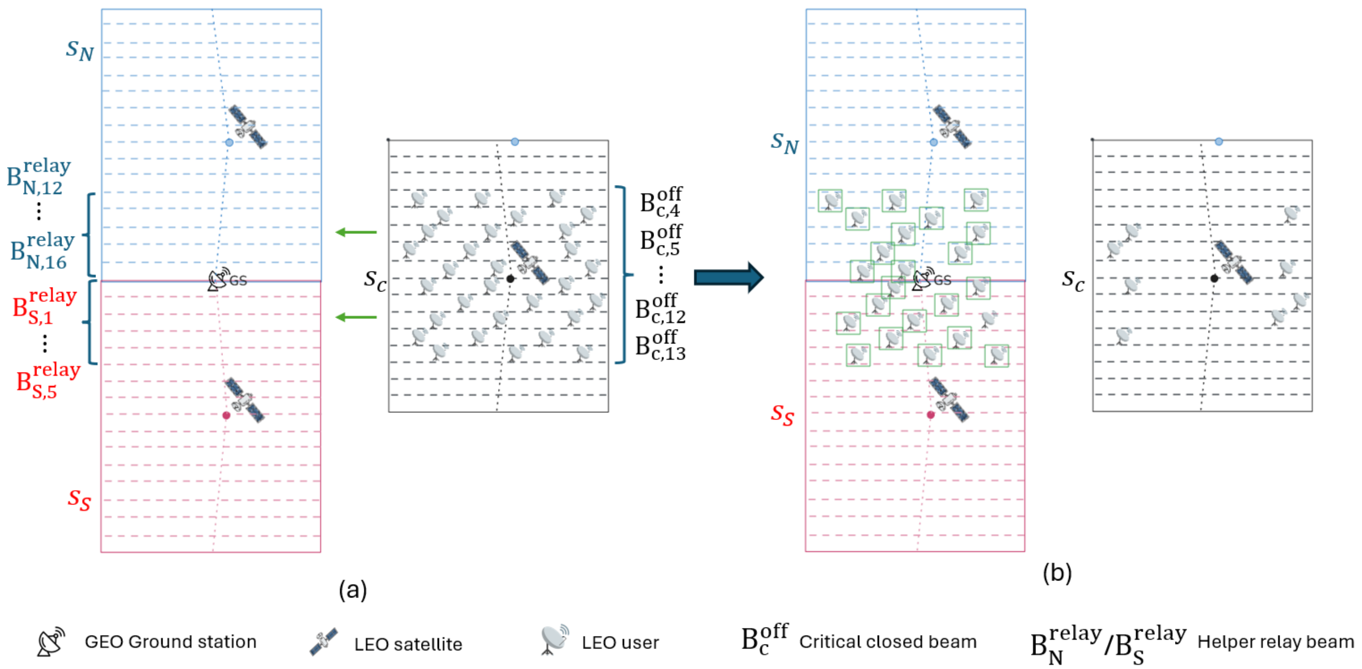


Fig. 6. Illustration of Beam-Pair Local Relay (BPLR). Candidate critical users in closed beams are reassociated to the corresponding helper relay beams according to $\mathcal{M}_{\text{off}\to\text{relay}}(t)$.

Let K_t^{SAHR} denote the number of candidate helper users considered in SAHR at time slot t, and let K_t^{BPLR} denote the number of candidate closed-beam users considered in BPLR. In the SAHR stage, candidate construction and released-power computation are linear in K_t^{SAHR}, while sorting the candidates by released power dominates the complexity:

$$\mathcal{O}\big(K_t^{\text{SAHR}}\log K_t^{\text{SAHR}}\big).$$

In the relay-beam boosting stage, let H_t denote the number of helper relay beams in time slot t. Sorting helper relay beams by relay demand requires

$$\mathcal{O}\big(H_t\log H_t\big).$$

In the BPLR stage, candidate users are sorted by achievable relay service rate, and then allocated sequentially according to the remaining relay capacity. The dominant complexity is therefore

$$\mathcal{O}\big(K_t^{\text{BPLR}}\log K_t^{\text{BPLR}}\big).$$

Over all time slots, the online complexity is

$$\mathcal{O}\Big(\sum_{t=1}^{T}\big(K_t^{\text{SAHR}}\log K_t^{\text{SAHR}}+H_t\log H_t \\ +K_t^{\text{BPLR}}\log K_t^{\text{BPLR}}\big)\Big).$$

Thus, SAPR-R reduces the online decision process to local sorting and sequential allocation within precomputed beam relations, instead of solving a global association and continuous power-control problem in every time slot.

H. Summary

This chapter presented SAPR-R for maintaining user service under the GS EPFD constraint during critical time slots. The method first precomputes time-slot beam relation records from predictable satellite geometry, beam footprints, and beam-level EPFD information. During online operation, SAPR-R applies SAHR, helper relay beam boosting, and BPLR in sequence.

The staged design separates the original coupled problem into local decisions. SAHR uses critical safe beams to take over selected helper users and release helper satellite power. Helper relay beam boosting allocates the available relay power budget according to relay demand. BPLR then reassociates closed-beam users to feasible helper relay beams within predefined beam pairs. In this way, SAPR-R improves user satisfaction during critical periods while satisfying the EPFD constraint.

V. Evaluation

A. Evaluation Overview

This chapter evaluates the performance of the proposed Safe-Beam Assisted Power-Release Relay (SAPR-R) method, focusing on its ability to improve user satisfaction and service continuity under the aggregate EPFD constraint. The evaluation consists of three main parts. First, we observe the aggregate EPFD variation during the critical satellite pass over the GS, as well as the number of beams that must be shut down to satisfy the EPFD constraint. This analysis confirms the existence of EPFD violation and beam shutdown demand during the critical pass, and further supports the rationale of designing a relay-based recovery method. Second, we compare different interference mitigation methods in terms of service performance and relay capability. Finally, we further analyze the impact of different EPFD limits on user satisfaction to evaluate the stability and applicability of SAPR-R under different interference constraint conditions.

B. Simulation Setup

To evaluate the performance of the proposed SAPR-R method under the NGSO–GSO coexistence scenario, we design a simulation framework that combines STK and MATLAB. This framework is used to construct the LEO constellation, obtain the time-varying satellite geometry, and further evaluate the performance of different interference mitigation methods in terms of EPFD compliance, downlink capacity, and user satisfaction.

The simulation uses the following two main tools:

- **STK (Satellite Tool Kit):** [19] STK is used to construct the OneWeb-like LEO constellation and generate the relative positions, visibility relationships, and beam coverage information among the LEO satellites, the GS, and users at each time slot. Based on the system information generated by STK, the dynamic geometric variation during the critical satellite pass near the GS can be simulated.
- **MATLAB:** [20] MATLAB is used to process the system information generated by STK and perform offline preprocessing. This includes constructing the beam–user association, critical/helper satellite relationship, and critical-beam/helper-beam mappings. MATLAB is then used to execute the proposed SAPR-R method and calculate the EPFD and user satisfaction performance of each method at different time slots.

The main simulation parameters used in this chapter are summarized in Table I. The table integrates the settings related to the LEO constellation, GSO satellite, GS, LEO user ground station, and EPFD evaluation. These parameters are used in the following EPFD evaluation and user satisfaction comparison.

C. Compared Schemes

To evaluate the proposed SAPR-R method, we compare it with several interference mitigation methods. All methods are evaluated under the same simulation parameters to ensure a fair comparison. The compared schemes are described as follows:

- **Beam shutdown only:** When the aggregate EPFD exceeds the limit, the high-interference beams on the critical satellite are directly shut down to satisfy the EPFD constraint. This method does not perform relay.
- **PC + Tilt:** This method represents a power-control and satellite-tilting based interference mitigation strategy. It reduces the interference toward the GS by adjusting the transmit power and satellite tilt angle. Although this method can help satisfy the EPFD constraint, it does not provide a relay-based recovery mechanism for users.
- **Only BPLR:** This method only applies the BPLR stage. Helper relay beams serve users located under the critical closed beams using the default power of the helper satellite. The SAHR stage is not applied.
- **SAPR-R (Proposed):** This method includes both the SAHR and BPLR stages.

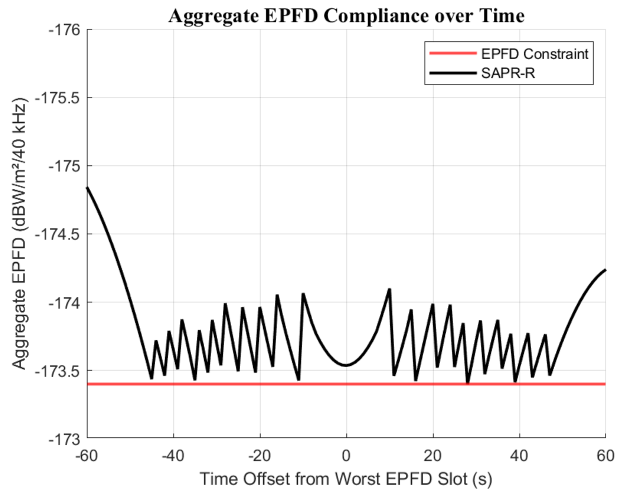


Fig. 7. Aggregate EPFD over time during the critical pass.

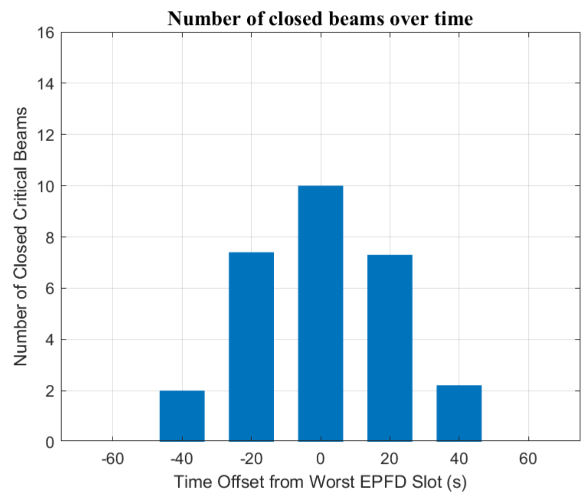


Fig. 8. Number of closed beams over time during the critical pass.

D. Evaluation Results

Based on the simulation setup and compared schemes described above, this section evaluates the performance of SAPR-R during the critical pass. The analysis is divided into four parts. First, the aggregate EPFD and the number of closed beams are examined to verify whether SAPR-R can still satisfy the EPFD constraint after relay-based recovery, and to observe the beam shutdown behavior of the critical satellite at different time slots. Second, the average satisfaction of different methods is compared under different user loads to analyze how increasing system load affects service performance. Third, the number of relayed critical users is used to explain the main reason why SAPR-R improves satisfaction, and the satisfaction CDF is further used to observe the satisfaction distribution of all users. Finally, the average user satisfaction of SAPR-R under different EPFD limits is evaluated to observe how the strictness of the interference limit affects service performance.

1) EPFD Compliance and Beam Shutdown:

This subsection first analyzes the aggregate EPFD after

TABLE I
SIMULATION PARAMETERS.

Parameter	Value
LEO constellation	
Altitude (km)	1200
Orbit plane inclination	$87.9°$
Time slot	1 s
Maximum transmit power	16.8 W per satellite
Maximum antenna gain	34.33 dBi
Downlink frequency	11.7 GHz
Bandwidth	250 MHz
Antenna pattern	[14] $A\exp(\beta\phi)$
Maximum tilt angle	$10°$
Approximated antenna gain coefficient, (A)	4.0×10^3
Approximated antenna gain coefficient, (B)	-0.0671
GSO satellite	
Altitude (km)	35785.4
Longitude	30.6
Latitude	0
Downlink frequency	11.7 GHz
GSO ground station	
Minimum elevation angle	$10°$
Maximum receive gain	38.1 dBi
Noise temperature	240 K
Antenna pattern	[8]
LEO user ground station	
Maximum receive gain	40.95 dBi
Noise temperature	240 K
Antenna pattern	[8]
User demand	50 Mbps
EPFD	
EPFD limit (dB(W/m^2/1 MHz))	-173.4 [5]
Reference antenna diameter	60 cm

applying SAPR-R and further observes the number of beams that must be shut down on the critical satellite to satisfy the EPFD constraint. This analysis is used to verify whether the proposed method can still satisfy the EPFD limit after relay-based recovery, and to show the beam shutdown behavior during the critical pass.

Fig. 7 shows the aggregate EPFD variation during the critical satellite pass near the GS. The red line represents the EPFD constraint, and the black line represents the aggregate EPFD after applying SAPR-R. As shown in the figure, SAPR-R keeps the aggregate EPFD below the limit during the entire observation period. This indicates that the proposed method does not cause EPFD violation while performing service recovery for closed-beam users. Therefore, SAPR-R can compensate for the service loss caused by beam shutdown while maintaining EPFD compliance.

Fig. 8 shows the number of beams that must be shut down on the critical satellite at different time offsets. When the satellite approaches the worst EPFD slot, the number of closed beams gradually increases and reaches its maximum around the worst EPFD slot. After the satellite moves away from GS, the number of closed beams gradually decreases. This result shows that the beam shutdown demand changes with the satellite–GS geometry.

2) User Satisfaction under Different User Loads:

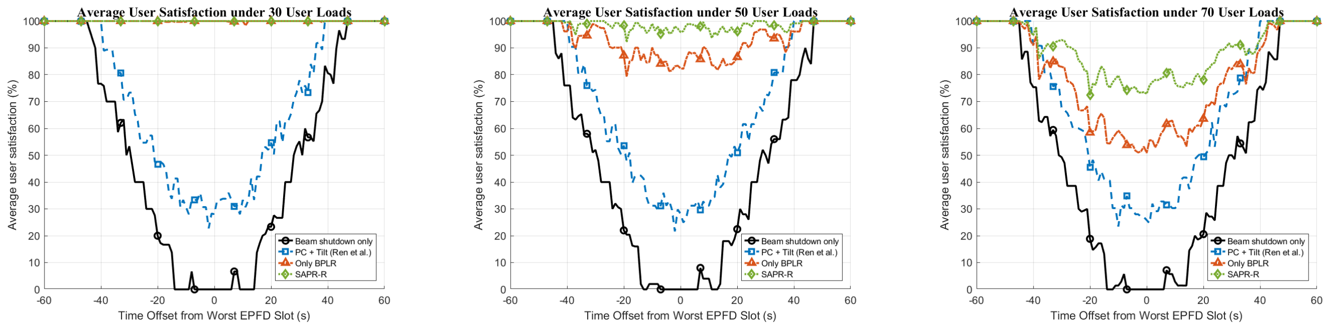
This subsection compares the average satisfaction of different interference mitigation methods under different user loads. By changing the number of users, we can observe how the service performance changes as the system load increases, and evaluate whether SAPR-R can maintain user satisfaction under different load conditions.

Fig. 9 shows the average satisfaction under three user load scenarios: 30 users, 50 users, and 70 users. As shown in the figure, Beam shutdown only has the lowest satisfaction because it does not provide relay service for users affected by closed beams. PC + Tilt can reduce the interference toward the GS, but it does not recover the service of users affected by beam shutdown. Only BPLR can serve part of the closed-beam users through helper relay beams, so it achieves better satisfaction than Beam shutdown only and PC + Tilt.

Compared with the other methods, SAPR-R maintains higher average satisfaction under all three user loads. When the user load is low, the helper satellite usually has enough available power, so the difference between Only BPLR and SAPR-R is small. However, as the user load increases, more closed-beam users need relay service, and Only BPLR becomes limited by the default power of the helper satellite. In contrast, SAPR-R uses SAHR to release available power from the helper satellite and improve the relay capability of helper relay beams. Therefore, SAPR-R can maintain more stable service performance under medium and high user loads.

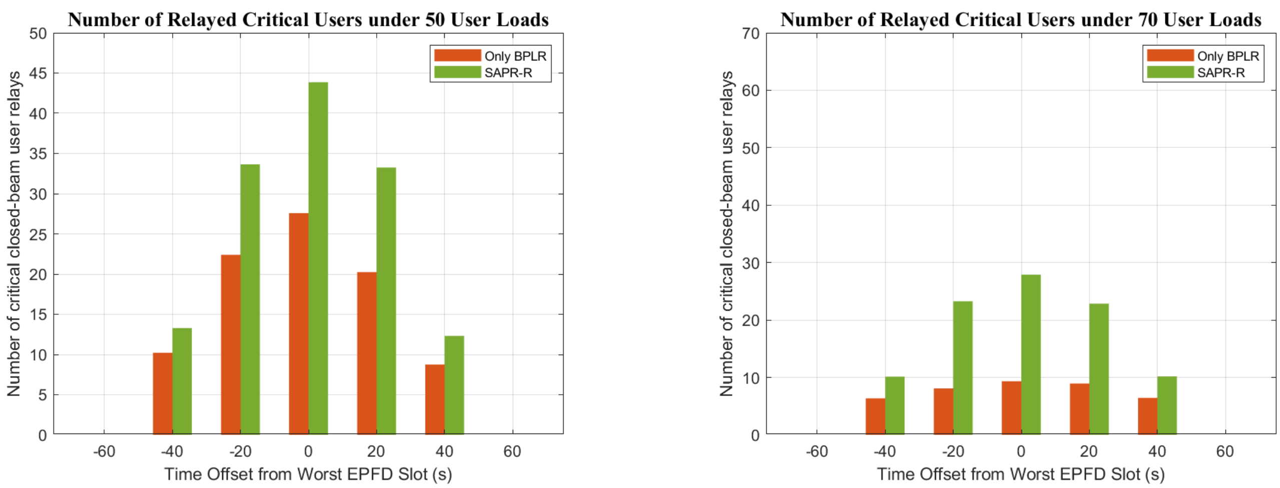
3) Relay Capability and Satisfaction Distribution:

This subsection further analyzes why SAPR-R improves user satisfaction and observes the satisfaction distribution of different methods. Since 30 users is a low-load scenario, the default available power of the helper satellite is usually sufficient to support most relay demand. Therefore, the difference in relay capability is not obvious under 30 users. To better show the effect of the proposed algorithm, this subsection focuses on the 50-user and 70-user scenarios.



(a) 30 users. (b) 50 users. (c) 70 users.

Fig. 9. Average satisfaction under different user loads. (a) 30 users. (b) 50 users. (c) 70 users.



(a) 50 users. (b) 70 users.

Fig. 10. Number of relayed critical users with and without SAHR. (a) 50 users. (b) 70 users.

Fig. 10 compares the number of successfully relayed critical users between Only BPLR and SAPR-R under the 50-user and 70-user scenarios. As shown in the figure, SAPR-R serves more critical users in both load scenarios. The difference becomes more obvious around the worst EPFD slot, where the beam shutdown demand is the most severe. This result shows the necessity of Stage 1 in SAPR-R.

Fig. 11 shows the satisfaction CDF of three interference mitigation methods. This figure focuses on the user-level satisfaction distribution. A CDF curve that shifts further to the right means that more users maintain higher satisfaction. In contrast, if the curve rises quickly in the low-satisfaction region, more users suffer from service degradation.

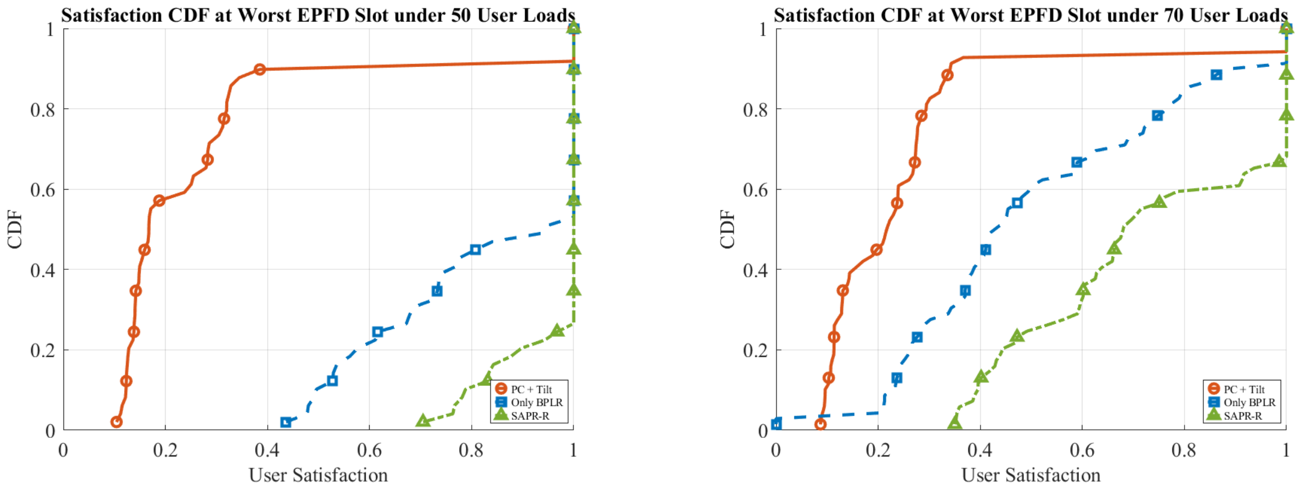
As shown in the figure, the CDF curve of SAPR-R shifts more toward the high-satisfaction region than the other methods. This means that more users can maintain better service quality. The result shows that SAPR-R not only improves average satisfaction, but also reduces the proportion of low-satisfaction users, making the overall user service quality more stable.

4) Impact of Different EPFD Limits:

This subsection further evaluates the impact of different EPFD limits on the service performance of SAPR-R. When the EPFD limit becomes stricter, the allowed aggregate interference becomes lower. As a result, the critical satellite may need to shut down more beams to satisfy the constraint, and the relay demand of closed-beam users may also increase. This analysis is used to observe whether SAPR-R can maintain stable average user satisfaction under different EPFD limits.

Fig. 12 shows the average user satisfaction of SAPR-R under different EPFD limits. As shown in the figure, when the EPFD limit is more relaxed, SAPR-R can maintain higher average user satisfaction. This result shows that the strictness of the EPFD limit directly affects user satisfaction. It also indicates that SAPR-R can achieve better service performance under more relaxed interference constraints.

Overall, SAPR-R keeps the aggregate EPFD below the limit during the critical pass and uses SAHR to improve the ability of helper relay beams to serve critical users. Compared with other interference mitigation methods, SAPR-R maintains higher average satisfaction under different user loads and reduces the proportion of low-satisfaction users. In addition, the results under different EPFD limits show that the strictness of the interference constraint affects user satisfaction, and SAPR-R can maintain better service performance under more relaxed constraints. Overall, SAPR-R achieves a better balance



(a) 50 users. (b) 70 users.

Fig. 11. CDF of user satisfaction under different interference mitigation methods.

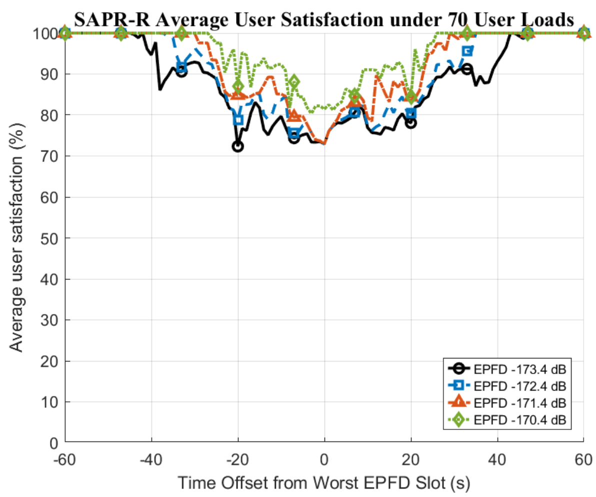


Fig. 12. Average user satisfaction of SAPR-R under different EPFD limits.

between EPFD compliance and service continuity.

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